

SOA Survey

Morphologically Grounded ECG Arrhythmia Classification: A CNN-BiLSTM-Attention Pipeline with P - QRS - T Segment Overlap for Clinical XAI

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Abstract

Automated ECG arrhythmia classification has matured rapidly since 2017, yet a fundamental gap blocks clinical adoption: state-of-the-art 1D convolutional networks achieve high benchmark accuracy by exploiting global signal statistics and spectral shortcuts rather than the localised morphological landmark, P -wave shape, QRS width, ST -segment slope, that a cardiologist inspects beat by beat. This survey reviews 10 papers spanning architectural surveys of deep ECG methods [1] to anatomy-aware augmentation [3] and prototype-based explainability [6]. We identify a tractable, precisely bounded research gap: *standard 1D-CNNs are not morphologically grounded*, meaning their attention and gradient saliency maps do not reliably localise to the P - QRS - T regions that define a given arrhythmia. We therefore propose a five-week build of a **CNN-BiLSTM-Attention** pipeline on Lead II recordings drawn from the PTB-XL Diagnostic superclass, evaluated by two complementary metrics: macro- F_1 -score and a P - QRS - T Attention Overlap score (AO) that measures whether model attention falls within Pan-Tompkins-detected P -wave or QRS boundaries. The pipeline is implemented using NeuroKit2 [8] for preprocessing, PyTorch for the neural backbone, and Captum [10] for Grad-CAM and Integrated Gradients. Morphological grounding, not raw accuracy, is the primary success criterion.

(AF) is characterised by the absence of organised P -waves and an irregularly irregular RR -interval. A cardiologist detects each condition through a sequential, landmark-driven process: inspect the P -wave, measure the PR interval, assess QRS morphology, then the ST -segment and QT interval.

State-of-the-art deep networks do *not* reason this way. As documented by Ansari et al. [1] across 368 studies, benchmark macro- F_1 -scores are frequently driven by global texture shortcuts such as spectral envelopes, amplitude statistics, and recording-level demographic co-variables, rather than the specific morphological intervals that carry diagnostic meaning. Three compounding failure modes arise:

Failure Mode 1: Shortcut Learning. Models trained on unbalanced data associate the dominant texture of the majority class (Normal Sinus Rhythm, >50% of PTB-XL) with all low-confidence inputs. Gradient attribution maps then localise to the QRS complex, the highest-amplitude segment, even when the decisive feature is the *absent* P -wave (AF) or the *elevated* ST -segment (ischemia).

Failure Mode 2: Augmentation-Morphology Conflict. Standard augmentation (Gaussian noise, global time-warping, random cropping) is applied without awareness of the cardiac cycle. A global time-warp stretching a 10-second window by 5% simultaneously and incorrectly elongates the PR interval, QRS duration, and QT interval, generating a Normal record that superficially mimics LQTS [3].

Failure Mode 3: Physiologically Naive Denoising. Butterworth bandpass filters (0.5–40 Hz) atten-

1. Problem Definition

1.1. The Shortcut-Learning Crisis in Cardiac AI

The 12-lead electrocardiogram (ECG) is the non-invasive gold standard for detecting cardiac arrhythmias. Diagnostically critical conditions impose strict morphological signatures: Wolff-Parkinson-White (WPW) syndrome produces a δ -wave preceding the QRS complex; Long QT syndrome (LQTS) requires a QT interval exceeding 480 ms; atrial fibrillation

uate the low-frequency *P*-wave envelope while suppressing powerline interference at the cost of *R*-peak sharpness. A model whose training data has had its *P*-waves smoothed away cannot learn *P*-wave-dependent diagnoses.

1.2. Proposed Solution: Morphological Grounding via CNN-BiLSTM-Attention

This project addresses the gap with a **modular pipeline** that enforces morphological grounding at each stage. The pipeline has three stages:

1. **Physiological Preprocessing (SWT + NeuroKit2):** Stationary Wavelet Transform (SWT) denoising preserves *R*-peak sharpness and *P*-wave amplitude; Pan-Tompkins via NeuroKit2 provides beat segmentation boundaries.
2. **Anatomy-Aware Augmentation (STAR):** Beat-wise sinusoidal resampling simulates heart rate variability without distorting the *P-QRS-T* ordering within each beat.
3. **CNN-BiLSTM with Single Attention Head:** A 1D CNN extracts beat-level morphological features; a BiLSTM captures inter-beat rhythm context; a single-head attention mechanism produces a timestep-weighted context vector whose weights are the explanation.

2. Literature Review

2.1. Deep Learning for ECG: Progress and the Morphology Gap

Ansari et al. [1] survey 368 deep learning studies for ECG arrhythmia detection from 2017 to 2023. 1D CNNs dominate ($\approx 59\%$ of studies), with MIT-BIH used in 61% of cases. Crucially, only 28% of studies applied any augmentation, and *none* evaluated whether augmentation preserved the *P-QRS-T* structure that underlies clinical reasoning. This establishes that the morphology gap is systemic, not incidental, and motivates the grounding objective of this project.

Talukder et al. [2] present an end-to-end framework combining a 1D-CNN with Grad-CAM and SHAP across six arrhythmia classes on MIT-BIH and PTB-XL. Their key finding provides the calibration refer-

ence for our AO metric: Grad-CAM maps localise to the *QRS* complex for ventricular arrhythmias and to the *P*-wave region for supraventricular classes. A model whose attention weights replicate this localisation pattern has achieved a degree of clinical reasoning that post-hoc saliency maps can only approximate.

The xECGArch dual-branch architecture of Hammer et al. [4] separates a short-term branch (0.6 s windows, morphology) from a long-term branch (10 s windows, rhythm), validated via pixel-flipping. This architectural separation of morphology and rhythm is the design principle this project inherits and simplifies into a sequential CNN-BiLSTM-Attention pipeline that a single student can implement and debug on a single GPU within one week.

The ECG-XPLAIM model of Pantelidis et al. [5] provides multi-scale Inception-style 1D-CNN baselines and open-source Grad-CAM code, serving as the post-hoc XAI control against which our attention-based AO is compared in Week 4.

2.2. SWT Denoising for Morphology Preservation

Wavelet-based denoising resolves the bandwidth trade-off of Butterworth filters through multi-resolution analysis. The Discrete Wavelet Transform (DWT) with a Daubechies-8 (db8) wavelet decomposes signal $x(t)$ into detail coefficients $\{d_j\}$ and an approximation $\{a_J\}$ across J levels:

$$x(t) = a_J(t) + \sum_{j=1}^J d_j(t). \quad (1)$$

Baseline wander (< 0.5 Hz) concentrates in a_J and can be zeroed independently; powerline artefact (50 Hz) falls in d_1 and is soft-thresholded without touching the *P*-wave or *R*-peak coefficients. The Stationary Wavelet Transform (SWT) is preferred because it is shift-invariant, avoiding the sub-sampling artefacts that introduce spurious *R*-peak jitter during DWT reconstruction [11]. Implementation uses `pywt.swt` from PyWavelets [12].

2.3. Anatomy-Aware Augmentation: STAR

Sinusoidal Time-Amplitude Resampling (STAR) [3] is designed explicitly as a physiological simulator.

For each RR -segment i , a sinusoidal coefficient $c_i \in (1-\alpha, 1+\alpha)$ governs both time-axis resampling and proportional amplitude scaling:

$$x'_i(t) = c_i \cdot x_i\left(\frac{t}{c_i}\right), \quad t \in [R_i, R_{i+1}]. \quad (2)$$

Because scaling is *intra-beat* and *proportional*, the P - QRS - T ordering is preserved within every beat. STAR has been shown to yield statistically significant F_1 -score improvements over Gaussian noise, random cropping, and global time-warping, with gains attributed to the preservation of morphological invariance [3]. This makes STAR suitable as the sole augmentation strategy without requiring complex generative model training.

2.4. Explainability: From Saliency to Case-Based Reasoning

Captum [10] provides a unified PyPI-installable library implementing Grad-CAM, Integrated Gradients, and Layer Conductance for arbitrary PyTorch models. For 1D-CNN and BiLSTM-Attention architectures, Integrated Gradients computes the expected gradient of the output class score with respect to the input signal, integrated along a straight path from a zero-baseline:

$$\text{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha, \quad (3)$$

where x' is the baseline (zero signal) and F is the model output. Integrated Gradients satisfy completeness and sensitivity axioms, making them preferable to vanilla gradient maps for time-series [10].

ProtoECGNet [6] establishes the intrinsic-XAI upper bound: predictions are made by learned similarity to prototype beat exemplars constrained to correspond to actual training beats, yielding explanations of the form “classified as AF because it most resembles a prototype showing absent P -wave.” This defines the AO ceiling our attention-based model aspires to approach. Implementing ProtoECGNet is designated as an optional stretch goal in Week 5.

3. Research Gap and Proposed Direction

3.1. The Morphological Grounding Gap

The surveyed literature establishes a precise, actionable gap: *standard 1D-CNN pipelines for ECG classification are not morphologically grounded*. Gradient and attention maps from these models routinely localise to the QRS complex, the globally dominant amplitude feature, regardless of the predicted condition. For conditions defined by P -wave absence (AF), ST -segment elevation (ischemia), or pre- QRS delta wave (WPW), a model that highlights only the QRS complex is clinically uninformative despite high benchmark F_1 .

The gap has three jointly unaddressed dimensions:

1. **Denoising and Morphology:** No deep learning study has demonstrated that SWT-denoised training data measurably improves attention localisation to P -wave or ST -segment regions compared to Butterworth-filtered baselines, evaluated with a structured segment-overlap metric.
2. **Augmentation and Architecture:** STAR [3] is validated in isolation; no work jointly trains a CNN-BiLSTM-Attention model on STAR-augmented data and measures whether the combination improves morphological attention localisation.
3. **Architecture and Explanation:** Single-head attention in a BiLSTM encoder produces timestep-weight vectors that are naturally interpretable as saliency maps, yet no pipeline has calibrated these weights against Pan-Tompkins-derived P/QRS segment boundaries.

Formal Research Question. *Does a CNN-BiLSTM-Attention model trained on SWT-denoised, STAR-augmented Lead II recordings achieve statistically significantly higher P -QRS-T Attention Overlap (AO) than a 1D-CNN baseline trained on Butterworth-filtered data, confirming that physiological preprocessing and anatomy-aware augmentation together improve morphological grounding beyond what architecture alone contributes?*

3.2. The P-QRS-T Attention Overlap Metric

The *P-QRS-T Attention Overlap* (AO) is the primary explanation-quality metric:

$$AO_k = \frac{|\mathcal{T}_k \cap \mathcal{R}_{\text{seg}}|}{|\mathcal{R}_{\text{seg}}|}, \quad (4)$$

where $\mathcal{T}_k$ is the set of top- k attention-weighted (or top- k Grad-CAM-activated) timesteps, and $\mathcal{R}_{\text{seg}}$ is the set of timesteps within Pan-Tompkins-detected *P*-wave or *QRS* complex boundaries for the predicted condition. For AF, $\mathcal{R}_{\text{seg}}$ is the inter-*QRS* *P*-wave window; for ventricular arrhythmias, it is the *QRS* complex; for ischemia, it is the *ST*-segment (*QRS*-offset to *T*-wave onset). A model that correctly highlights the *P*-wave absence window for AF achieves $AO_k \approx 1$; a model that defaults to *QRS* highlighting for every class achieves $AO_k \approx 0$ for *P*-wave-dependent conditions.

4. Benchmark, Evaluation, and Data Landscape

Dataset and Subset Selection. The full PTB-XL dataset (21,837 recordings, 12-lead, 500 Hz) is the canonical multi-label ECG benchmark. For our project, we restrict to **Lead II** and the **Diagnostic superclass**, yielding five coarse categories: Normal (NORM), Myocardial Infarction (MI), ST/T-Change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). This reduces the preprocessing surface from 12×5000 to 1×5000 samples per recording, halves GPU memory requirements, and confines the morphological analysis vocabulary to the well-characterised Lead II waveform that is diagnostically standard for rhythm and inferior ischemia assessment.

Table 1: Dataset used in this project.

Dataset	Size	Subset	Notes
PTB-XL	21,837	Lead II, Diagnostic	5 super-classes; primary corpus.
MIT-BIH	48 rec.	2-lead	OOD rhythm stress test only.

Evaluation Protocol. A strict *inter-patient split* is enforced: no patient’s recordings appear in both training and test sets. The canonical split scripts from the PTB-XL benchmarking suite [7] are used directly. Primary classification metrics are **macro- F_1 -score** and **macro-AUPRC**; raw *Accuracy* is not reported as a primary result because the Normal class dominates (>50% of records). The primary explanation-quality metric is AO_k (Equation 4) for $k \in \{50, 100, 150\}$ timesteps, reported as mean $\pm$ standard deviation across test records per Diagnostic superclass.

Compute Context. All experiments target a single consumer GPU (8 GB VRAM or above; tested on NVIDIA RTX 3060). Training on the Lead II Diagnostic subset is estimated to require <2 hours per run, leaving adequate time within each week for debugging and ablation.

Experimental Design Considerations

Week 1: Data Cleaning and Lead II Extraction

Goal: Build and validate the physiologically faithful data preparation layer.

- Download PTB-XL (21,837 records, 500 Hz). Filter to the Diagnostic superclass using the `superdiagnostic_cd` column. Extract Lead II (index 1) from each 12-lead recording. Apply the inter-patient split using the `strat_fold` column from the benchmarking suite [7].
- Implement SWT denoising via PyWavelets (`pywt.swt`, `db8`, $J = 8$ levels; Equation 1). Zero the approximation coefficient a_8 (baseline wander). Apply SURE soft-thresholding to d_1 (50 Hz powerline). Reconstruct via inverse SWT.
- **Morphological fidelity check:** Run Pan-Tompkins *R*-peak detection via NeuroKit2 (`nk.ecg_peaks`) [8] on both raw and SWT-denoised signals. Confirm: (i) *R*-peak shift <2 ms; (ii) mean *P*-wave amplitude (80–200 ms pre-*R*-peak window) retained >95%; (iii) *R*-peak sharpness (maximum absolute first derivative in a 5-sample window) retained >90%.
- Use NeuroKit2 (`nk.ecg_delineate`) to extract

P -wave onset/offset and QRS onset/offset timestamps. These constitute the $\mathcal{R}_{\text{seg}}$ ground truth for AO computation in Week 4.

- **Deliverable:** Frozen `.pt` tensors for the SWT-denoised split (training, validation, test) and a paired Butterworth-filtered control split, both with identical class-count distributions. A quality-check notebook confirming all three morphological fidelity criteria.

Week 2: STAR Augmentation and CNN-BiLSTM Backbone Implementation

Goal: Implement anatomy-aware augmentation and the modular neural backbone.

- **STAR augmentation:** Apply Equation 2 with $\alpha = 0.15$ to minority classes (MI, STTC, CD, HYP) to reach approximately 50% of the Normal sample count per class. After each augmentation, re-run `nk.ecg_delineate` and confirm that P - QRS - T landmark count and ordering are preserved; reject and resample any beat that fails this check.
- **CNN-BiLSTM-Attention architecture:** The backbone is a three-module sequential stack.

Module 1 (Morphology CNN): Four 1D convolutional blocks with kernel sizes $\{3, 7, 15, 31\}$ (doubling channels from 32 to 256), BatchNorm, ReLU, and max-pooling. Input: single Lead II beat segment of 256 samples zero-padded at R -peak boundaries. Output: feature map $\mathbf{F} \in \mathbb{R}^{T' \times 256}$.

Module 2 (Rhythm BiLSTM): A 2-layer Bidirectional LSTM (hidden size 128 per direction; `torch.nn.LSTM` with `bidirectional=True`) encodes the CNN feature map sequence:

$$\mathbf{H} = [\overrightarrow{\text{LSTM}}(\mathbf{F}); \overleftarrow{\text{LSTM}}(\mathbf{F})] \in \mathbb{R}^{T' \times 256}. \quad (5)$$

Module 3 (Single-Head Attention): A learnable query vector $\mathbf{v} \in \mathbb{R}^{256}$ produces scalar timestep weights:

$$\alpha_t = \frac{\exp(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_{t'}))}, \quad (6)$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t. \quad (7)$$

The context vector $\mathbf{c}$ feeds a two-layer MLP classification head with sigmoid output for multi-label prediction. The weight vector α is the intrinsic saliency map used in Week 4.

- **Deliverable:** Unit-tested PyTorch implementation with verified forward-pass tensor shapes. A vanilla 1D-CNN baseline (same CNN backbone, global average pooling, no BiLSTM or attention) as the ablation stub.

Week 3: Model Training and Hyperparameter Tuning

Goal: Train the full CNN-BiLSTM-Attention pipeline on the physiologically faithful data split and establish classification baselines.

- Train the CNN-BiLSTM-Attention model on: (a) the SWT + STAR-augmented split (*proposed*), and (b) the Butterworth + Gaussian-noise-augmented split (*control*). Train the vanilla 1D-CNN baseline on both splits for ablation.
- Use class-weighted binary cross-entropy loss, AdamW optimiser ($\text{lr} = 1 \times 10^{-3}$, weight decay 10^{-4}), and a cosine annealing schedule over 50 epochs with 5-epoch linear warmup.
- Confirm strict inter-patient split via a patient-ID intersection check between training and test tensors before the first training run.
- Report macro- F_1 , macro-AUPRC, and per-class F_1 for all five Diagnostic superclasses. Compare proposed vs. control and BiLSTM-Attention vs. CNN-only to isolate the contribution of each module.
- **Deliverable:** Four trained checkpoints (`.pt` files) and a primary classification results table. Validation loss curves confirming convergence without overfitting.

Week 4: Grad-CAM, Attention Maps, and Segment Overlap Measurement

Goal: Measure whether the pipeline activates on the clinically correct morphological segment.

- **Attention saliency:** Extract the timestep weight vector α (Equation 6) from the CNN-BiLSTM-

Attention model for each test record. Upsample α to the original 500 Hz sample resolution via linear interpolation.

- **Grad-CAM saliency:** Using Captum [10] (`captum.attr.LayerGradCam`), extract Grad-CAM maps from the final convolutional layer of the vanilla 1D-CNN baseline. Use Captum’s `IntegratedGradients` as a second attribution method (Equation 3) for both models.
- **P-QRS-T Attention Overlap:** Load *P/QRS* boundary timestamps computed in Week 1 via `NeuroKit2`. Compute AO_k (Equation 4) for $k \in \{50, 100, 150\}$ across all test records. Report mean $\pm$ standard deviation per Diagnostic superclass for each combination of (model $\times$ saliency method).
- **Shortcut detection test:** Inject synthetic *ST*-elevation artefacts (a Gaussian bump of amplitude 0.3 mV, duration 40 ms) into 50 Normal Sinus Rhythm test records. A model exploiting a global amplitude shortcut will shift its saliency to the injected feature ($AO_k \approx 0$ by design for the non-ischemic condition); a morphologically grounded model should not.
- **Deliverable:** AO_k comparison table (proposed vs. control, attention vs. Grad-CAM vs. Integrated Gradients) and a heatmap figure overlaying saliency weights on raw Lead II ECG traces for at least one representative beat per superclass.

Week 5: Comparative Analysis and Report

Goal: Compile the final comparative analysis and produce the submission-ready report.

- Assemble the primary results table comparing: (a) proposed CNN-BiLSTM-Attention (SWT + STAR) vs. (b) 1D-CNN control (Butterworth + Gaussian noise), across macro- F_1 , macro-AUPRC, and AO_k for each superclass.
- Perform a paired Student’s *t*-test on per-record AO_k scores (proposed vs. control) to confirm statistical significance at $p < 0.05$.
- **Ablation summary:** Quantify the individual contributions of (i) SWT vs. Butterworth denoising,

(ii) STAR vs. Gaussian augmentation, and (iii) BiLSTM + attention vs. global average pooling, by holding all other factors constant.

- **Stretch goal (optional):** Run the official Pro-toECGNet [6] repository on the same SWT-denoised test split to establish the intrinsic-XAI AO ceiling. Compare prototype similarity scores against attention weights as saliency sources.
- **Report formatting:** Compile all figures, tables, and equations into the final $\text{\LaTeX}$ report. Confirm that all cited repositories are accessible, all cited DOIs resolve, and all `.pt` checkpoints are archived.
- **Deliverable:** Submission-ready PDF report, archived model checkpoints, and a reproducibility script (`reproduce.sh`) executing the full pipeline from raw PTB-XL download to final AO_k table.
- **Evaluation pitfalls to avoid:** *Data leakage:* enforce inter-patient split with patient-ID intersection check; *Metric gaming:* report macro- F_1 as primary, not raw *Accuracy*; *Attention instability:* run each model with three random seeds and report AO_k standard deviation across seeds; *Denoising over-correction:* confirm *P*-wave amplitude retention $>95\%$ before training.

5. Annotated Reading List

References

- [1] Y. Ansari, O. Mourad, K. Qaraqe, and E. Serpedin. Deep learning for ECG arrhythmia detection and classification: an overview of progress for period 2017–2023. *Frontiers in Physiology*, 14:1246746, 2023. DOI: <https://doi.org/10.3389/fphys.2023.1246746> *What it does:* Systematically surveys 368 deep learning studies for ECG arrhythmia detection, categorising architectures by dataset, preprocessing, and evaluation metric. *Why it is here:* Confirms that 1D CNNs dominate, only 28% of studies apply augmentation, and none evaluate augmentation’s

- effect on explanation quality. Establishes the morphological grounding gap as systemic.
- [2] Md A. Talukder, A. S. Talaat, N. J. Muna, A. Alazab, M. Kazi, and U. K. Das. An explainable deep learning framework for trustworthy arrhythmia detection from ECG signals. *Scientific Reports*, 15:39496, 2025. DOI: <https://doi.org/10.1038/s41598-025-22986-0> *What it does*: End-to-end 1D-CNN with Grad-CAM and SHAP across six arrhythmia classes on MIT-BIH and PTB-XL. *Why it is here*: Provides the calibration reference for AO: Grad-CAM localises to the *QRS* for ventricular arrhythmias and to the *P*-wave region for supraventricular classes. The direct Grad-CAM baseline for the Week 4 comparison.
 - [3] Nader Nemati. Comparative Analysis of Data Augmentation Techniques for Clinical ECG Classification with Sinusoidal Time–Amplitude Resampling (STAR). *PMC*, 2025. PMCID: PMC12606187. *What it does*: Introduces STAR, a beat-wise augmentation applying sinusoidal time-warping and amplitude scaling within each *RR* interval, preserving *P-QRS-T* structure while simulating HRV. *Why it is here*: The sole augmentation strategy for Week 2; canonical reference for Equation 2.
 - [4] A. Hammer, M. Gottling, H. Malberg and A. Linke. Fusion of automatically learned rhythm and morphology features matches diagnostic criteria in ECG arrhythmia classification. *Scientific Reports*, 2024. *What it does*: Introduces xECGArch, a dual-branch CNN separating 0.6 s morphology windows from 10 s rhythm windows, with pixel-flipping evaluation. *Why it is here*: Architectural precedent for morphology-rhythm separation; provides the pixel-flipping shortcut detection protocol adapted in Week 4.
 - [5] P. Pantelidis et al. ECG-XPLAIM: eXplainable Locally-adaptive Artificial Intelligence Model for arrhythmia detection from large-scale electrocardiogram data *medRxiv*, 2025. *What it does*: Multi-scale Inception-1D CNN with open-source Grad-CAM overlays and pre-trained PTB-XL weights. *Why it is here*: Provides the post-hoc Grad-CAM baseline against which attention-based AO is compared in Week 4.
 - [6] S. Sethi et al. ProtoECGNet: Case-Based Interpretable Deep Learning for Multi-Label ECG Classification with Contrastive Learning. *Machine Learning for Healthcare (MLHC)*, 2025. *What it does*: Prototype-based architecture where predictions are made by similarity to learned beat exemplars, providing intrinsic case-based explanations. *Why it is here*: Defines the intrinsic-XAI AO ceiling in Week 5 (optional stretch goal).
 - [7] N. Strodthoff, P. Wagner, T. Schaeffter, and W. Samek. Deep learning for ECG analysis: Benchmarks and insights from PTB-XL. *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. DOI: <https://doi.org/10.1109/JBHI.2020.3022989> *What it does*: Introduces the PTB-XL benchmarking suite with canonical inter-patient split scripts, multi-label baselines (ResNet-1D, Inception-1D, xResNet), and macro- F_1 /AUPRC evaluation utilities. *Why it is here*: The definitive reference for the inter-patient split and evaluation protocol used in Weeks 3–5.
 - [8] D. Makowski, T. Pham, Z. J. Lau, J. C. Brammer, F. Lespinasse, H. Pham, C. Schölzel, and S. H. A. Chen. NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods*, 53(4):1689–1696, 2021. DOI: <https://doi.org/10.3758/s13428-020-01516-y> *What it does*: Open-source Python library implementing Pan-Tompkins *R*-peak detection (`nk.ecg_peaks`), *P/QRS/T* delineation (`nk.ecg_delineate`), and a comprehensive suite of ECG preprocessing utilities. *Why it is here*: Core preprocessing tool in Week 1 for segment boundary extraction; provides the $\mathcal{R}_{\text{seg}}$ ground truth for AO computation.
 - [9] J. Pan and W. J. Tompkins. A real-time QRS detection algorithm. *IEEE Transactions on Biomedical Engineering*, 32(3):230–236, 1985. DOI: <https://doi.org/10.1109/TBME.1985.325532> *What it does*:

- Introduces the Pan-Tompkins algorithm: band-pass filtering, differentiation, squaring, moving-window integration, and adaptive thresholding for real-time R -peak detection. *Why it is here:* Foundational reference for the R -peak detection step that partitions beats and defines $\mathcal{R}_{\text{seg}}$ boundaries in Weeks 1 and 4.
- [10] N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson. Captum: A unified and generic model interpretability library for PyTorch. *arXiv preprint*, arXiv:2009.07896, 2020. URL: <https://arxiv.org/abs/2009.07896> *What it does:* Implements Integrated Gradients, Layer Grad-CAM, Layer Conductance, and 15+ other attribution methods for arbitrary PyTorch models under a unified Attribution interface. Install: `pip install captum`. *Why it is here:* The attribution library for Week 4; LayerGradCam provides the 1D-CNN baseline saliency map and IntegratedGradients provides the model-agnostic attribution map used in AO computation.
- [11] P. S. Addison. *The Illustrated Wavelet Transform Handbook: Introductory Theory and Applications in Science, Engineering, Medicine and Finance*. Taylor & Francis, 2005. *What it does:* Provides the mathematical foundation for DWT/SWT decomposition and the clinical rationale for db8 wavelet selection in biomedical signal processing. *Why it is here:* Reference for Equation 1 and the morphological fidelity arguments in Week 1.
- [12] G. R. Lee, R. Gommers, F. Wasilewski, K. Wohlfahrt, and A. O’Leary. PyWavelets: A modern Python wavelet transforms package. *Journal of Open Source Software*, 4(36):1237, 2019. DOI: <https://doi.org/10.21105/joss.01237> *What it does:* Official Python library implementing `pywt.swt` (shift-invariant Stationary Wavelet Transform) with db8 basis, SURE thresholding utilities, and inverse reconstruction. Install: `pip install PyWavelets`. *Why it is here:* The SWT denoising engine for Week 1; canonical reference for the `pywt.swt` calls in the preprocessing pipeline.
- [13] D. Tymoshchuk, O. Semeniv, and R. Tkachenko. AutoML PyCaret and SHAP Explainable AI for ECG Signal Classification on Micro-controllers. *CEUR Workshop Proceedings*, 2024/2025. *What it does:* Demonstrates >99% *Accuracy* on a simple 5-class ECG task using AutoML amplitude features. Illustrates that raw *Accuracy* on class-imbalanced datasets is an unreliable metric. *Why it is here:* Motivates the use of macro- F_1 and macro-AUPRC over *Accuracy* in the proposed evaluation protocol.

6. Preliminary Implementation Resources

The repositories below cover every core component of the proposed pipeline. Each is active, publicly accessible, and directly installable via `pip` or `git` clone.

- <https://github.com/neuropsychology/NeuroKit> **NeuroKit2: ECG preprocessing and P/QRS/T delineation.** Official repository implementing Pan-Tompkins R -peak detection (`nk.ecg_peaks`), P -wave onset/offset and QRS boundary extraction (`nk.ecg_delineate`), and signal quality indices. Used in Week 1 to produce the $\mathcal{R}_{\text{seg}}$ boundary tensors required for AO computation. Install: `pip install neurokit2`.
- https://github.com/helme/ecg_ptb-xl-benchmarking **PTB-XL benchmarking suite.** Canonical inter-patient split scripts, macro- F_1 and macro-AUPRC evaluation utilities, multi-label baselines (ResNet-1D, Inception-1D, xResNet), and PTB-XL data loaders. Used in Weeks 1–3 for the split, class-weighted sampling, and baseline F_1 comparison.
- <https://github.com/PyWavelets/pywt> **PyWavelets: SWT denoising engine.** Implements `pywt.swt` with db8 wavelet, soft-thresholding utilities, and inverse SWT reconstruction. Core preprocessing module for Week 1. Install: `pip install PyWavelets`.

- https://github.com/DeepPSP/torch_ecg_torch_ecg **1D-CNN ECG baselines in PyTorch.** A comprehensive, actively maintained library of ECG-specific PyTorch modules including ResNet-1D, Multi-Scale CNN, and CRNN architectures with PTB-XL data loaders. Used in Weeks 2–3 to initialise the CNN backbone and as a reference implementation for the vanilla 1D-CNN ablation baseline. Install: `pip install torch-ecg`.
- <https://github.com/pytorch/captum> **Captum: Grad-CAM and Integrated Gradients.** Official PyTorch interpretability library. `captum.attr.LayerGradCam` extracts Grad-CAM maps from the final convolutional layer of the 1D-CNN baseline; `captum.attr.IntegratedGradients` computes Integrated Gradients (Equation 3) for both models. Used in Week 4 for all saliency extraction and AO computation. Install: `pip install captum`.
- <https://github.com/antonior92/automatic-ecg-diagnosis> **ResNet-1D: Validated CNN backbone.** Well-tested 1D ResNet with SE blocks, class-weighted loss, inter-patient split scripts, and evaluation utilities. Provides a second single-branch CNN ablation baseline and a reference implementation confirming that the CNN module in the proposed pipeline is correctly implemented (Weeks 2–3).
- <https://github.com/ppantele/ECG-XPLAIM> **ECG-XPLAIM: Multi-scale Inception-1D baseline with Grad-CAM.** Open-source multi-scale Inception-style 1D-CNN with pre-trained PTB-XL weights and a Grad-CAM reference implementation. Serves as the post-hoc XAI baseline in Week 4 against which attention-based AO scores are directly compared.
- <https://github.com/bbj-lab/protoecgnet> **ProtoECGNet: Optional intrinsic-XAI upper bound.** Official repository for prototype-based multi-label ECG classification on PTB-XL. Used in Week 5 (stretch goal) to establish the AO ceiling achievable by an intrinsically explainable architecture. Pre-trained weights and contrastive pretraining hooks are included.
- <https://github.com/pytorch/pytorch> **PyTorch: BiLSTM and attention implementation.** `torch.nn.LSTM` (with `bidirectional=True`) implements the rhythm BiLSTM branch (Equation 5). The single-head attention module (Equations 6–7) is implemented as a lightweight `nn.Module` using standard linear layers.